University of Toronto
Combustion Processes (AER315)
Instructor: Rajat Sawanni
Handout #7
November 16, 2020

Turbulent Combustion

Outline *In this handout we finally see the structure and working of flames in turbulent flows. Turbulence as a problem still remains elusive to the scientific community but it doesn't stop us from using the advantageous effects of turbulence in practical combustion devices. In this handout, we first look at some of the defining features of turbulence and the problems encountered in its solution (especially the closure problem). This part of the module is not required but is offered for your understanding of turbulence. Following the same, we study the presence of chemical reactions in such turbulent flows, again characterising them on the basis of mixing i.e. premixed and non-premixed turbulent flames. We note the distinguished features of such flames and present the models and theories existing in such conditions. Finally, a scaling of important flame parameters are studied based on our understanding and limitations.*

Contents

1 Turbulence - Basics

1.1 Introduction

The existence and smoothness of solution for the Navier Stokes equation is still an unsolved problem in physics and a million dollar one at that. Assuming that the hypothesis holds and there exists a smooth solution for Navier Stokes with a bounded kinetic energy, the equation is notoriously hard to solve under turbulent conditions. Unlike laminar flows where the fluid motion across different layers of the flow is parallel, or flowing in "lamina", turbulent flows are unsteady, irregular, seemingly random and chaotic. We shall see that such flows do not allow a strict analytical study, and one usually depend on intuition and dimensional arguments. Inspite of our everyday experience with it, it is extremely hard to even precisely define *turbulence*. Instead, we rely on some of the characteristics of turbulence to essentially identify when we see it. Some of these characteristics are as follows:

- *Randomness:* Turbulent flows seem irregular, unpredictable and chaotic. A chaotic system in the dynamic sense implies that a small change in the initial conditions can result in a completely different solution after time t. The random unsteadiness associated with various flow properties remains the hallmark of a turbulent flow and is illustrated for the axial velocity component in Fig. 1

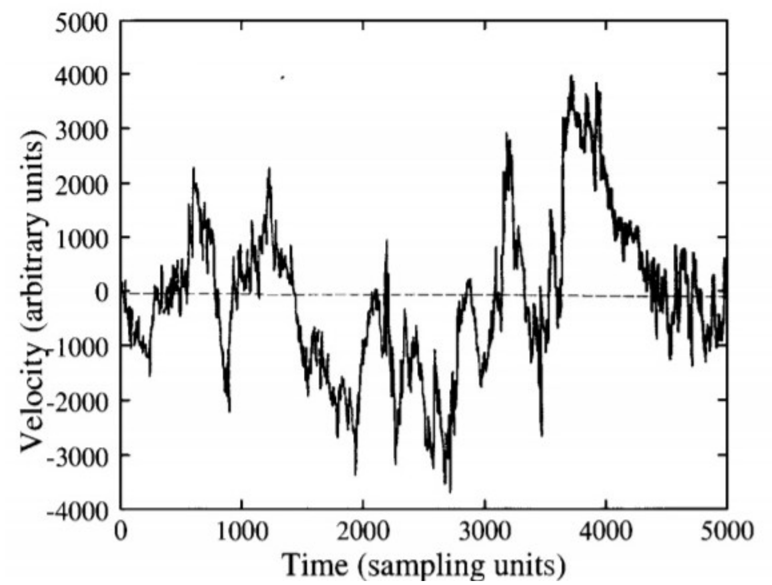


Figure 1: Velocity as a function of time at a fixed point in a turbulent flow.

- *Non-linearity:* Turbulent flows are highly nonlinear. This non-linearity serves 2 important functions: a) as a bed for growing instability and b) in facilitating the energy cascade. Turbulent flow usually results when small perturbations in the velocity field causes instabilities in the flow and are not sufficiently damped by viscous action. We are all familiar with the experiment of Osborne Reynolds [1] in which the transition from a laminar to a turbulent flow in a tube was visualized with a dye streak and related to the dimensionless parameter that now bears the name of this important early fluid mechanician, i.e., the Reynolds number.

The Reynolds number is a dimensionless parameter which represents the ratio of inertial and viscous forces in a fluid. At sufficiently low Re values, the viscous forces are large and small perturbations in the flow are dampened to the effect that a smooth laminar flow is seen. With an increase in the inertial forces, the perturbations in non-linear inertial terms grow further and a transition to turbulent flow occurs. The properties of a transitional turbulence is marked with a departure from flow symmetries and fluctuations in the flow properties. As the turbulent flow further develops at much higher Re, it transitions to fully developed turbulence and appears to have a tendency to restore the symmetries in a statistical sense far from the boundaries. Turbulence at very high Reynolds numbers, when all or some of the possible symmetries are restored in a statistical sense, is known as *fully developed turbulence*. This breaking of symmetries as we transition to turbulence and its restoration is shown in Fig 2 where the flow past a cylinder is investigated at increasing Re.

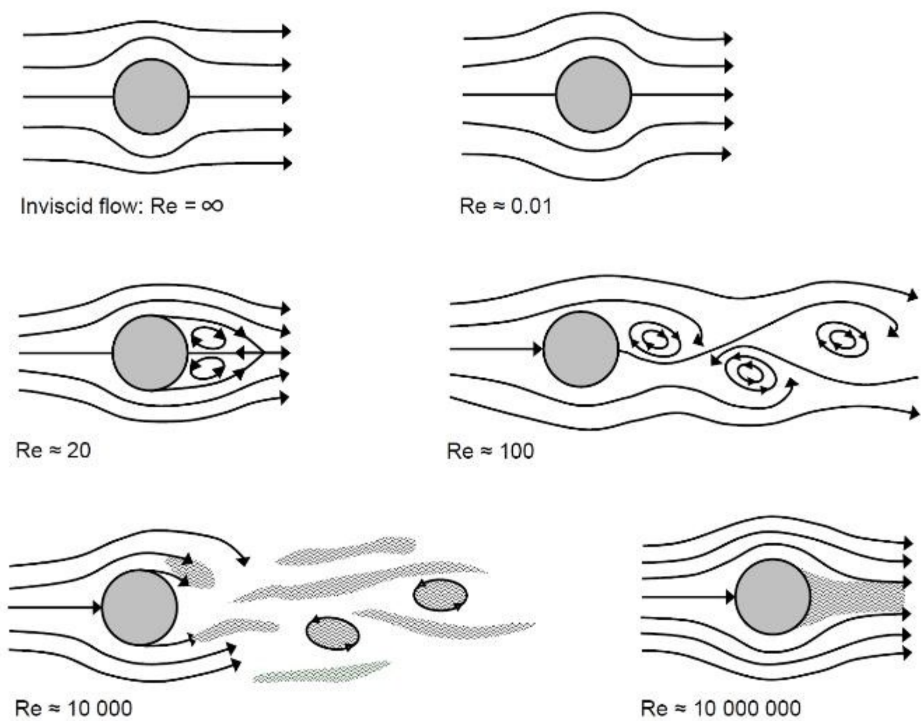


Figure 2: Flow around a cylinder for different Reynolds numbers (symmetries)

The second function of nonlinearity comes in the form of turbulence sustenance by energy cascading. Turbulent motion is generally characterised by motion in a variety of length scales. The large scale motion transfers its energy to the small scale motion (where viscous forces are considerable) and the small scale motion loose this energy to heat dissipation via viscous forces. This whole process is facilitated by the non-linearity of the turbulent flow dynamics. Consider for instance of turbulence generated by the air flow around a tall building: the energy-containing eddies generated by flow separation have sizes of the order of tens of meters. Somewhere downstream, dissipation by viscosity takes place, for the most part, in eddies at the Kolmogorov microscales: of the order of a millimetre for the present case. At the intermediate scales, there is neither a direct forcing of the flow nor a significant amount of viscous dissipation, but there is a net nonlinear transfer of energy from the large scales to the small scales.

- *Vorticity:* Turbulence is characterised by high levels of fluctuating vorticity. The identifiable structures in a turbulent flow are vaguely called eddies. Flow visualisation of turbulent flows shows various structures- coalescing, dividing, stretching and above all, spinning. A characteristic feature of turbulence is the existence of a multitude of eddy size ranges. The large eddies (integral length scales) have a size of the order of the width of the turbulent flow (ex. boundary layer thickness in case of boundary layer flows) and the smallest eddies are of the order of millimeters (Kolmogorov microscales). These eddies facilitate the energy cascading via the non-linear terms of the Navier Stokes equation. It would, thereby appear as if there is no way to get past viscous forces: as soon as the scale of the flowfield becomes so large that viscosity effects could conceivably be neglected, the flow creates small scale motion, thus keeping viscosity effects (in particular dissipation rates) at a finite level.
- *Diffusivity:* Due to the macroscopic mixing of fluid particles, turbulent flows are characterised by a rapid rate of diffusion and mixing of momentum and heat. This is also one of the most important reason for its use in combustion related applications.

1.2 Mathematical Description

Since actual computation of the raw flow properties is not possible in high Re flows, the standard analysis of turbulence separates fluctuating property from its time-mean property. The time average of any given quantity Q is given by:

$$\overline{Q} = \frac{1}{T}\int_{t_0}^{t_0+T} Q dt \tag{1}$$

where T is sufficiently large than any period of fluctuations in Q. The mean value $\overline{Q}$ itself may vary slowly with time, as sketched in Fig.3, and we speak of this case as an unsteady turbulent flow. We may, then define the fluctuating component of the velocity, Q' as what remains when the mean flow is subtracted.

$$Q' = Q - \overline{Q} \tag{2}$$

By definition, the mean of fluctuating velocity is zero, i.e. $\overline{Q'} = 0$. Therefore, to characterise the fluctuation, we work with its mean-square value, i.e.

$$\overline{Q'^2} = \frac{1}{T}\int_{t_0}^{t_0+T} Q'^2 dt \tag{3}$$

The root mean square value of Q' is defined as $Q'_{rms} = (\overline{Q'^2})^{1/2}$. If the integrals is (1),(2) and (3) are independent of starting time t_0, then the fluctuations are said to be statistically stationary.

Although statistical theory and numerical simulation are viable options, most of the research on turbulent-flow analysis in the past century has used the concept of time averaging. Applying time averaging to the basic equations of motion yields the Reynolds equations, which involve both mean and fluctuating quantities. One then attempts to model the fluctuation terms by relating them

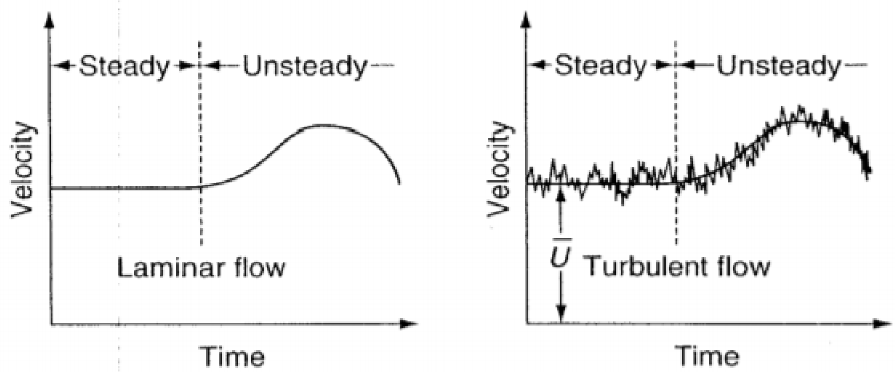


Figure 3: Steady and Unsteady laminar and turbulent flow

to mean properties or their gradients. This approach may now be yielding diminishing returns: Lumley (1989) gives a stimulating discussion of how time averaging might outlive its usefulness. The Reynolds equations are far from obsolete, however, and form the basis of most engineering analyses of turbulent flow. Let us now consider only incompressible turbulent flow with constant transport properties but with possible significant fluctuations in velocity, pressure, and temperature:

$$\begin{aligned} &u = \overline{u} + u' \qquad v = \overline{v} + v' \qquad w = \overline{w} + w' \\ &p = \overline{p} + p' \qquad T = \overline{T} + T' \end{aligned} \tag{4}$$

Before substituting these functions into the basic equations, we can verify from the basic integral relation, (1), that the following rules of averaging apply for any two turbulent quantities f and g:

$$\begin{gathered} \overline{f'} = 0 \qquad \overline{\overline{f}} = \overline{f} \qquad \overline{f\overline{g}} = \overline{f}\,\overline{g} \\ \overline{f'\overline{g}} = 0 \qquad \overline{f+g} = \overline{f} + \overline{g} \qquad \overline{fg} = \overline{f'g'} + \overline{f}\,\overline{g} \\ \overline{\frac{\partial f}{\partial s}} = \frac{\partial \overline{f}}{\partial s} \qquad \overline{\int f ds} = \int \overline{f} ds \end{gathered} \tag{5}$$

Now consider the incompressible continuity equation

$$\nabla . V = 0 \tag{6}$$

substitute in u, v, and w from (4), and take the time average of the entire equation. The result is

$$\frac{\partial \overline{u}}{\partial x} + \frac{\partial \overline{v}}{\partial y} + \frac{\partial \overline{w}}{\partial z} = 0 \tag{7}$$

Now subtract (7) from (6) but do not take the time average. This gives

$$\frac{\partial u'}{\partial x} + \frac{\partial v'}{\partial y} + \frac{\partial w'}{\partial z} = 0 \tag{8}$$

Thus the mean and fluctuating velocity components each separately satisfy an equation of continuity. This would not be true if we accounted for density fluctuations, because terms involving

$\overline{\rho u_i'}$ would couple the two relations. Our interest will center on the mean equation (7), not the fluctuations (8). Now attempt the same procedure with the nonlinear Navier-Stokes equations.

$$\rho \frac{DV}{Dt} = \rho g - \nabla p + \mu \nabla^2 V \tag{9}$$

Before substituting for u, v, w, p, we can avoid unnecessary churning by noting the following rather clever rearrangement of the convective-acceleration term:

$$V.\nabla Q = u\frac{\partial Q}{\partial x} + v\frac{\partial Q}{\partial y} + w\frac{\partial Q}{\partial z} \equiv \frac{\partial}{\partial x}(uQ) + \frac{\partial}{\partial y}(vQ) + \frac{\partial}{\partial z}(wQ) \tag{10}$$

an identity which follows from the continuity relation, (6). If we now use this idea to substitute into (9) and take the time average, we obtain

$$\rho \frac{D\overline{V}}{Dt} + \rho \frac{\partial}{\partial x_j}(\overline{u_i' u_j'}) = \rho g - \nabla p + \mu \nabla^2 \overline{V} \tag{11}$$

Thus the mean momentum equation is complicated by a new term involving the turbulent inertia tensor $\overline{u_i' u_j'}$. This new term is never negligible in any turbulent flow and is the source of our analytic difficulties, because its analytic form is not known apriori. In essence, the time-averaging procedure has introduced nine new variables (the tensor components) which can be defined only through (unavailable) knowledge of the detailed turbulent structure. The components of $\overline{u_i' u_j'}$ are related not only to fluid physical properties but also to focal flow conditions (velocity, geometry, surface roughness, and upstream history), and no further physical laws are available to resolve this dilemma. In a two-dimensional turbulent boundary layer ($\overline{w} = 0, \partial/\partial z = 0$) the only significant term reduces to $\overline{u'v'}$, but even this single term requires extensive scratching about to achieve an analytic correlation which is semi-empirical at best. Some of the empirical approaches have been quite successful, though rather thinly formulated from non-rigorous postulates. A slight amount of illumination is thrown upon (11) if it is rearranged to display the turbulent inertia terms as if they were stresses, which of course they are not. Thus we write

$$\rho \frac{DV}{Dt} = \rho g - \nabla p + \nabla \cdot \tau_{ij} \tag{12}$$

where

$$\tau_{ij} = \mu \left(\frac{\partial u_i}{\partial x_j} + \frac{u_j}{x_i} \right) - \rho \overline{u_i' u_j'} \tag{13}$$

Mathematically, then, the turbulent inertia terms behave as if the total stress on the system were composed of the Newtonian viscous stresses plus an additional or apparent turbulent-stress tensor $-\rho\overline{u_i' u_j'}$. The turbulent stresses are still unknown, so that little has been gained but conceptual value. In a boundary layer, the dominant term : $-\rho\overline{u_i' u_j'}$ is called the turbulent shear.

Now consider the energy equation (first law of thermodynamics) for incompressible flow with constant properties

$$\rho c_p \frac{DT}{Dt} = k \nabla^2 T + \Phi \tag{14}$$

where Φ is the dissipation function. Taking the time-average, we obtain the mean-energy equation

$$\rho c_p \frac{D\overline{T}}{Dt} = -\frac{\partial}{\partial x_i}(q_i) + \overline{\Phi} \tag{15}$$

where

$$\overline{\Phi} = \frac{\mu}{2}\overline{\left(\frac{\partial \overline{u_i}}{\partial x_j} + \frac{\partial u_i'}{\partial x_j} + \frac{\partial \overline{u_j}}{\partial x_i} + \frac{\partial u_j'}{\partial x_i}\right)} \tag{16}$$

and

$$q_i = -k\frac{\partial \overline{T}}{\partial x_i} + \rho c_p \overline{u_i' T'} \tag{17}$$

By analogy with our rearrangement of the momentum equation, we have collected conduction and turbulent convection terms into a sort of total-heat-flux vector q, which includes molecular flux plus the turbulent flux $\rho c_v \overline{u_i' T'}$. The total-dissipation term $\overline{\Phi}$ is obviously complex in the general case. In two-dimensional turbulent-boundary-layer flow (the most common situation), the dissipation reduces approximately to

$$\Phi \approx \frac{\partial \overline{u}}{\partial y}\left(\mu \frac{\partial \overline{u}}{\partial y} - \rho \overline{u'v'}\right) \tag{18}$$

Equations (7), (12), and (15) are the Reynolds-averaged basic differential equations for turbulent mean continuity, mean momentum, and mean thermal energy. They contain many new unknowns involving time-mean correlations of fluctuating velocities and temperature. Therefore solutions cannot be attempted without additional relations or empirical modeling ideas.

1.3 Closure Problem

In two-dimensional turbulent-boundary-layer flow, the only additional unknown in the momentum equation is the turbulent shear $(-\rho\overline{u'v'})$. The traditional modeling assumption, following J. Boussinesq in 1877, is to make this a gradient diffusion term, analogous to molecular shear:

$$\tau_t = -\rho\overline{u'v'} = \mu_t \frac{\partial \overline{u}}{\partial y} \tag{19}$$

where μ_t is the eddy viscosity, which has the same dimension as μ but it is not a fluid property, varying instead with flow conditions and the geometry. (It depends upon the turbulent eddies, to put it colloquially.) In like manner we will define an eddy conductivity later for turbulent heat transfer

A very popular approach in this context is the mixing-length concept of Prandtl (1925), who, by analogy with kinetic theory, proposed that each turbulent fluctuation could be related to a length scale and a velocity gradient:

$$-\overline{u'v'} \approx (const)\left(l_1 \frac{\partial \overline{u}}{\partial y}\right)\left(l_2 \frac{\partial \overline{u}}{\partial y}\right) \tag{20}$$

where the scales l_1, and l_2, are called mixing lengths, which represent some mean eddy size much larger than the fluid's mean-free path. Now, for convenience, replace $(const)(l_1 l_2)$ by a single scale $\mathcal{L}^2$ a and compare Prandtl's idea with (20), The result is

$$\mu_t \approx \rho \mathcal{L}^2 \left| \frac{\partial \overline{u}}{\partial y} \right| \tag{21}$$

The model is complete if we can relate the mixing length $\mathcal{L}$ to the flow conditions. Different models have been suggested by authors depending on the flow boundary conditions. Other models apart from the mixing length model are also attempted and have been successful by involving composite functions for eddy viscosity. We call these as the zero equation models in turbulence modeling since it involves a direct implementation of eddy viscosity to the momentum equations. Apart from the zero equation model discussed above, additional equations for the solution to turbulence modeling may be classified as follows:

- One equation: usually a model of the turbulent-kinetic-energy equation.
- Two equations: the turbulent-kinetic-energy (K) relation, plus a model of either dissipation (ϵ), turbulence length scale (L), or vorticity fluctuations (ω). The first of these, the $K - \epsilon$ model, is probably the most popular scheme in use today.
- Reynolds stress models: these are computationally more complex and time-consuming but have the potential of greater accuracy and wider applicability.
- Almost model-free large-eddy simulation of turbulence.
- Model-free direct numerical simulation of turbulence.

2 Turbulent Premixed Flames

2.1 Flame Structure

One particular view of the structure of turbulent flames is illustrated in Fig. 12.6. Here we see superimposed instantaneous contours of convoluted thin reaction zones (Fig. 12.6a), obtained by visualizing the large temperature gradients in the flame using schlieren photography at different instants. The flow is from bottom to top with the flame stabilized above a tube through which the reactants flow into the open atmosphere. The instantaneous flame front is highly convoluted, with the largest folds near the top of the flame. The positions of the reaction zones move rapidly in space, producing a time-averaged view that gives the appearance of a thick reaction zone (Fig. 12.6b). This apparently thick reaction zone is frequently referred to as a turbulent flame brush. The instantaneous view, however, clearly shows the actual reaction front to be relatively thin, as in a laminar premixed flame. These reaction fronts are sometimes referred to as laminar flamelets.

Depending on the relative scales of turbulence and laminar flame thickness, the turbulent premixed flame front can take different characteristics. These characteristics are classified as regimes of turbulent premixed combustion and can be defined based on length scales as follows:

- $\delta_L \leq \eta$: *Wrinkled laminar flames:* When the flame thickness is smaller than the smallest scales of turbulent motions (Kolmogorov scales), the natural effect is that of the wrinkling of laminar flamelet by the turbulent motion. You can imagine a very thin string placed in

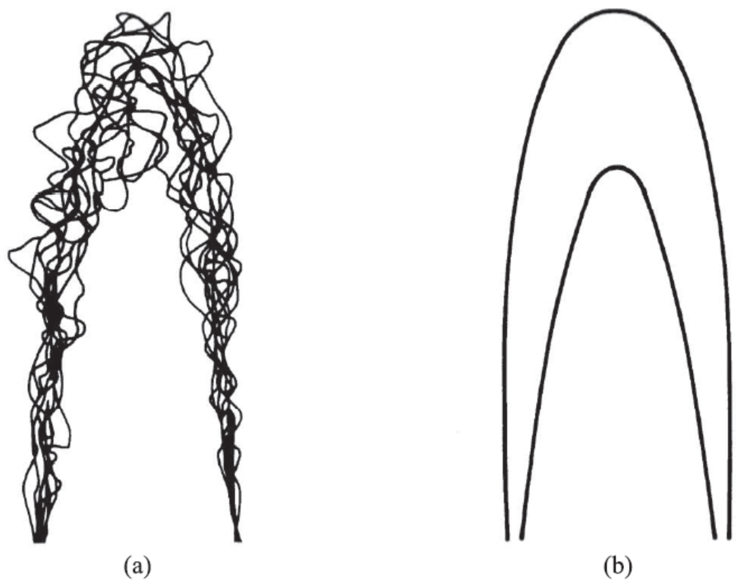


Figure 4: (a) Superposition of instantaneous reaction fronts obtained at different times. (b) Turbulent flame "brush" associated with a time-averaged view of the same flame.

turbulent motion undergoing rapid convolutions and wrinkling across its length. We can see in Fig.5 that reaction sheets are predicted only for Damköhler numbers greater than unity, depending on the turbulence Reynolds numbers, clearly indicating that the reaction-sheet regime is characterized by fast chemistry (in comparison with fluid mechanical mixing).

- $\delta_L > \Lambda$: *Distributed Reaction:* On the other hand, when flame thickness is greater than the largest scales of turbulent motion (integral scales), a distributed reaction zone is seen where the flame front is internally split by the turbulent motions and is spread across the whole flow-field. In this case, transport within the reaction zone is no longer governed solely by molecular processes, but is controlled, or at least influenced, by the turbulence. Fig.6a heuristically illustrates a distributed-reaction zone in which all turbulence length scales are within the reaction zone. Since reaction times are longer than eddy lifetimes ($Da < 1$), fluctuations in velocity, v_{rms}, temperature, T_{rms}, and species mass fractions, $Y_{i,rms}$, all occur simultaneously. Thus, the instantaneous chemical reaction rates depend on the instantaneous values of $T(=\overline{T} + T')$ and $Y_i(=\overline{Y_i} + Y_i')$, and fluctuate as well. Moreover, the time-average reaction rate is not simply evaluated using mean values, but involves correlations among the fluctuating quantities.

- $\Lambda > \delta_L > \eta$: *Flamelets in eddies*: This regime lies in between the two extremes and is marked by a scenario where the laminar flame thickness is larger than the smallest turbulence scales but smaller when compared to the largest scales. This implies that the turbulent eddies of the Kolmogorov scale perturb the flame front from inside and influence the transport within reaction zones. On the other hand, this perturbed flame front is then wrinkled by the large scale turbulence. This region is typified by moderate Damköhler numbers and high turbulence intensities. Fig.6b illustrates, conceptually, how combustion might proceed in this regime and

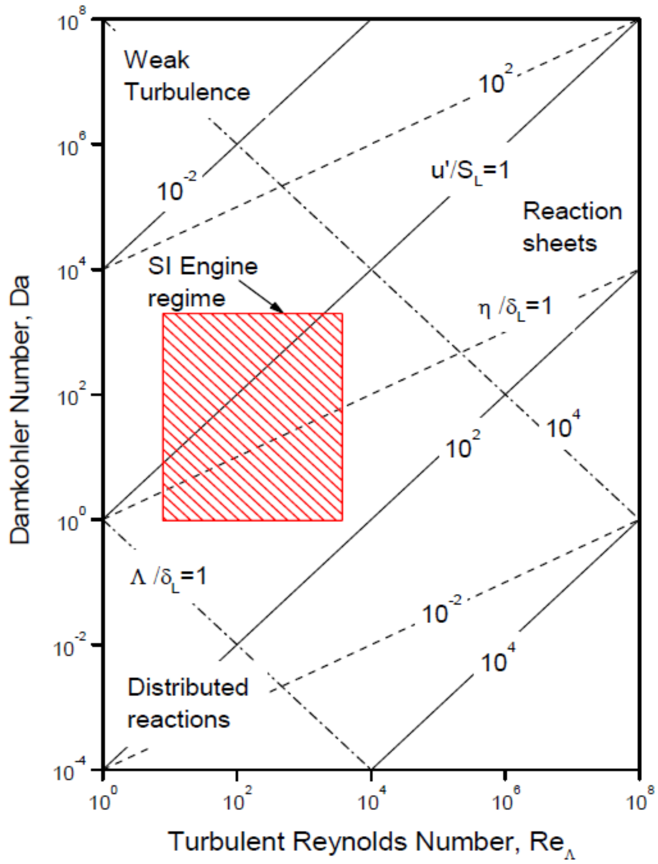


Figure 5: Important parameters characterizing turbulent premixed combustion.

follows the ideas that support the eddy-breakup model [4, 32]. As illustrated in Fig.6b, the burning zone consists of parcels of burned gas and almost fully burned gas. The intrinsic idea behind the eddy-breakup model is that the rate of combustion is determined by the rate at which parcels of unburned gas are broken down into smaller ones, such that there is sufficient interfacial area between the unburned mixture and hot gases to permit reaction

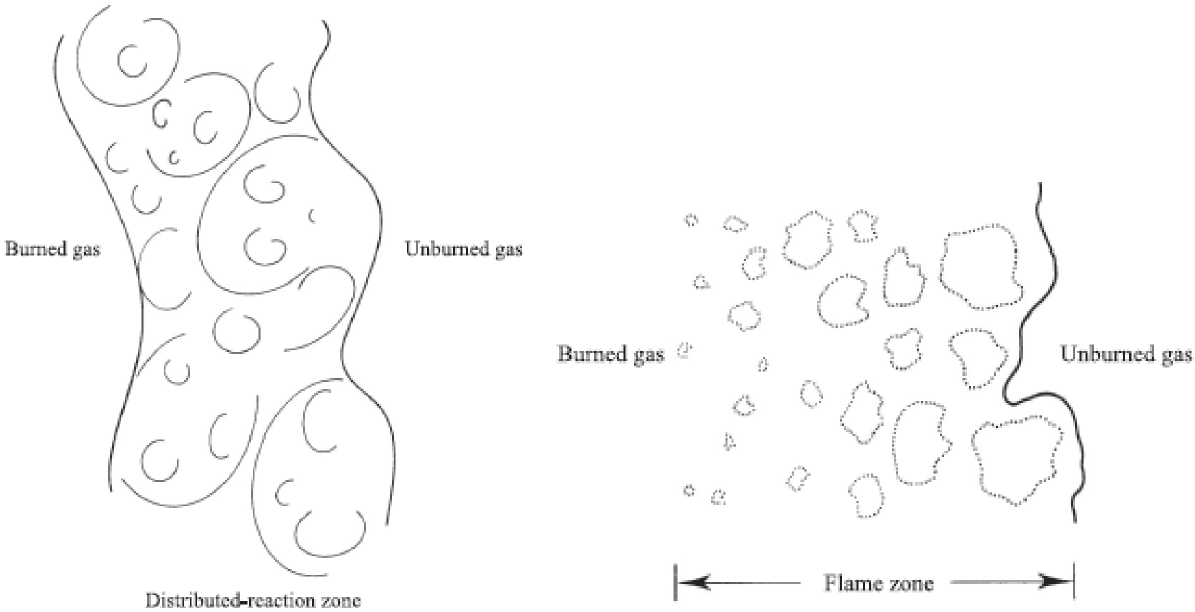
2.2 Definitions

- Laminar Flame thickness: as seen in previous lectures can be approximated as:

$$\delta_L \sim \alpha/S_L = \mathcal{D}/S_L = \nu/S_L \tag{22}$$

The above equality implies that we assumed (Schmidt Number) $Sc = \nu/\mathcal{D} = 1$, (Lewis Number) $Le = \alpha/\mathcal{D} = 1$, (Prandtl Number) $Pr = \nu/\mathcal{D} = 1$

- Turbulent Reynolds number:



(a) (b)

Figure 6: (a) Conceptual view of turbulent flame propagation in the distributed-reaction regime showing various length scales of turbulence within the reaction zone, (b) Conceptual view of turbulent flame propagation in the flamelets-in-eddies regime.

- based on Λ (integral length scale):

$$Re_\Lambda = \frac{u'\Lambda}{\nu} \tag{23}$$

- based on λ (Taylor microscale):

$$Re_\lambda = \frac{u'\lambda}{\nu} \tag{24}$$

- based on η (Kolmogorov length scale):

$$Re_\eta = \frac{u'\eta}{\nu} \tag{25}$$

Using the dependency of these length-scales on turbulent kinetic energy and dissipation terms, we can approximate:

$$Re_\Lambda \approx Re_\lambda^2 \approx Re_\eta^4 \tag{26}$$

- Turbulent Damköhler number: ratio of characteristic flow time, τ_{flow}, to the characteristic chemical time, τ_c

$$Da = \frac{\tau_{flow}}{\tau_c} \tag{27}$$

where the characteristic flow time in regards to the largest motion is given by $\tau_{flow} = \Lambda/u'$ and characteristic chemical time is given by $\tau_c = \delta_L/S_L$. Another expression for the Damköhler number is then given by:

$$Da = \frac{S_L \Lambda}{u' \delta_L} \tag{28}$$

- Karlovitz number:

$$Ka = \left(\frac{\Lambda}{\eta}\right)^2 = \frac{\delta_L u'}{S_L \lambda} \tag{29}$$

2.3 Turbulent Burning Velocity:

One of the most important unresolved problems in premixed turbulent combustion is the determination of the turbulent burning velocity. Above statement assumes that turbulent burning velocity is a well-defined quantity that only depends on local mean properties. However, there is no consensus in literature whether the turbulent burning velocity is a characteristic quantity that can be defined unambigously for different geometries.

Turbulent premixed flame propagation was first investigated by Damköhler (1940). He identified two limiting cases based on the magnitude of the scale of turbulence as compared to the thickness of the laminar premixed flame. For large scale turbulence, Damköhler assumed that the interaction between a turbulent premixed flame (wrinkled flame) front and the turbulent flame front is purely kinematic. Damköhler equated the mass flux $\dot{m}$ through the instantaneous turbulent flame surface area A_T with the mass flux through the cross-sectional area A_o. He used S_L for mass flux through A_T, and S_T for mass flux through A.

$$\dot{m} = \rho_u S_L A_T = \rho_u S_T A_o \tag{30}$$

$$\frac{S_T}{S_L} = \frac{A_T}{A_o} \tag{31}$$

Using geometric approximations, Damköhler proposed that (for wrinkled flame regime),

$$\frac{A_T}{A_o} = 1 + \frac{u'}{S_L} \tag{32}$$

In view of (31),

$$\frac{S_T}{S_L} = 1 + \frac{u'}{S_L} \tag{33}$$

where u' is the turbulent fluctuating velocity in the unburned gas.

Using similar geometric arguments, Schelkin showed that:

$$\frac{S_T}{S_L} = \left[1 + \left(\frac{2u'}{S_L}\right)^2\right]^{1/2} \tag{34}$$

Relationship proposed by Klimov:

$$\frac{S_T}{S_L} = 3.5\left(\frac{u'}{S_L}\right)^{0.7} \tag{35}$$

Clavin & Williams:

$$\frac{S_T}{S_L} = \left\{0.5\left[1+\left(1+8\frac{u'^2}{S_L^2}\right)^{1/2}\right]\right\}^{1/2} \tag{36}$$

Gülder:

$$\frac{S_T}{S_L} = 1 + 0.62\left(\frac{u'}{S_L}\right)^{1/2}\mathrm{Re}_\Lambda^{1/4} \tag{37}$$

For small-scale and high-intensity turbulence (Distributed Regime), Damköhler argued that turbulence only modifies the transport between the reaction zone and the unburned gas.

$$S_T/S_L \sim (D_T/D)^{1/2} \tag{38}$$

Since $D_T \propto u'\Lambda$ and $D \propto S_L\delta_L$. Then we have,

$$\frac{S_T}{S_L} \sim \left(\frac{u'\Lambda}{S_L\delta_L}\right)^{1/2} \tag{39}$$

For small-scale high-intensity turbulence conditions (usually called as distributed reaction regime), there are not many formulations available. In this regime, turbulent mixing is rapid as compared to the chemistry. For the distributed reaction regime the following semi-empirical model has been proposed, Gülder (1990):

$$\frac{S_T}{S_L} = 6.4\left(\frac{S_L}{u'}\right)^{3/4} \tag{40}$$

3 Turbulent Non-Premixed Flames

Historically, turbulent nonpremixed flames have been employed in the majority of practical combustion systems, principally because of the ease with which such flames can be controlled [1]. With current concern for pollutant emissions, however, this advantage becomes something of a liability in that there is also less ability to control, or tailor, the combustion process for low emissions.Because of the many applications of nonpremixed combustion, there are many types of nonpremixed flames. For example, simple jet flames are employed in glass melting and cement clinker operations, among others; unsteady, liquid-fuel sprays of many shapes are burned in diesel engines; flames stabilized by strong recirculation zones created by swirling flows or diverging walls are used in many systems, such as utility and industrial boilers; flames stabilized behind bluff bodies are used in turbojet afterburners; and partially premixed flames, created by nozzle-mix burners are employed in many industrial furnace. For any particular application, some of the issues the designer is faced with are as follows (importance of each may change depending on the nature of the application):

- Combustion intensity and efficiency
- Flame stability
- Flame shape and size
- Heat transport
- Pollutant emissions

3.1 Flame Structure

Turbulent non-premixed jet flames visually have brushy or fuzzy edges, similar to turbulent premixed flames; however, the non-premixed hydrocarbon flames generally are more luminous than their premixed counterparts, since some soot is usually present within the flame. Using diagnostics on such a flame reveals the intense reaction zones in the turbulent mixing pockets between the fuel jet and surrounding coflow air. The interaction of a turbulent jet creates eddies around the edges with a bigger motion scale and helps in effectively mixing the surrounding air to the central fuel jet. The luminosity at the base of the flame is quite weak, is blue in color, and is characteristic of a soot-free region. At higher levels in the flame, considerable quantities of soot exist, and the flame takes on a bright-yellow appearance. Fig. clearly shows the convoluted nature of the high-temperature region in which OH radicals are in abundance. We see that the width of the OH zones increase with downstream distance and the merging of the large high-temperature zones at the flame tip.

Figure 7: Instantaneous image of a turbulent non-premixed flame showing the higher sooting propensity as compared to premixed flames

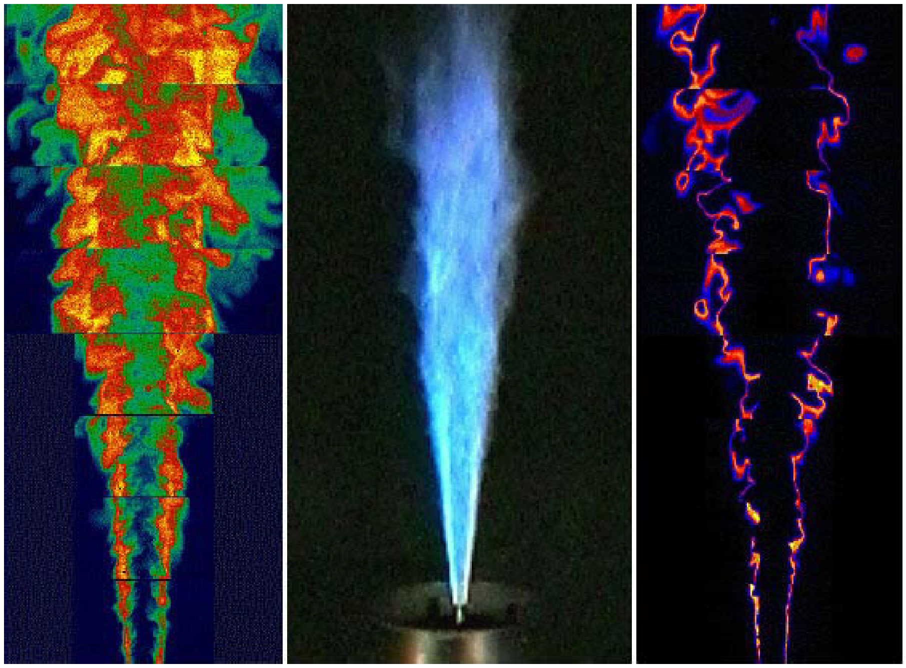


Figure 8: Images of a turbulent non-premixed flame with different diagnostics

3.2 Conserved Scalar Concept

The conserved scalar concept greatly simplifies the solution of reacting flow problems (i.e., the determination of the fields of velocity, species, and temperature), particularly those involving non-premixed flames. A conserved scalar is any scalar property that is conserved throughout the flowfield. For example, absolute enthalpy would qualify as being a conserved scalar if there are no sinks or sources of thermal energy. Element mass fractions are also conserved scalars. We will limit our discussion to only one, the mixture fraction:

$$f \equiv \frac{\text{Mass of material having its origin in the fuel stream}}{\text{Mass of mixture}} \tag{41}$$

Mixture fraction is uniquely related to Φ:

$$f \equiv \frac{\Phi}{(A/F)_{st} + \Phi} \tag{42}$$

For practical purposes we can define a flame boundary where $\Phi = 1$, then the mixture fraction at stoichiometric location:

$$f_s \equiv \frac{1}{(A/F)_{st} + 1} \tag{43}$$

For a three-species system, i.e., fuel, oxidizer, and products:

$$1 \text{ kg fuel} + (A/F) \text{ kg oxidizer} \longrightarrow [(A/F) + 1] \text{ kg products} \tag{44}$$

$$
\underbrace{f}_{\substack{\text{mass fraction of material}\\ \textit{having its origin}\\ \textit{in the fuel stream}}} = \underbrace{(1)}_{(\frac{\text{kg fuel stuff}}{\text{kg fuel}})} \underbrace{Y_F}_{(\frac{\text{kg fuel}}{\text{kg mixture}})} + \underbrace{\left(\frac{1}{(A/F)+1}\right)}_{(\frac{\text{kg fuel stuff}}{\text{kg products}})} \underbrace{Y_{Pr}}_{(\frac{\text{kg products}}{\text{kg mixture}})} + \underbrace{(0)}_{(\frac{\text{kg fuel stuff}}{\text{kg oxidizer}})} \underbrace{Y_{ox}}_{(\frac{\text{kg oxidizer}}{\text{kg mixture}})} \tag{45}
$$

fuel stuff is that material originating in the fuel stream. For a hydrocarbon fuel, fuel stuff is carbon and hydrogen. Another way of expressing mixture fraction is, then:

$$
f = Y_F + \left(\frac{1}{(A/F)+1}\right) Y_{Pr} \tag{46}
$$

Example: Consider a nonpremixed, ethane (C_2H_6) -air flame in which the mole fractions of the following species are measured using various techniques: C_2H_6, CO, CO_2, H_2, H_2O, N_2, O_2, and OH. The mole fractions of all other species are assumed to be negligible. Define a mixture fraction f expressed in terms of the mole fractions of the measured species. *Solution:* Approach will be to express f in terms of the known species mass fractions, and then express the mass fractions in terms of mole fractions.

$$
f \equiv \frac{\substack{\text{Mass of material having its}\\ \textit{origin in the fuel stream}}}{\text{Mass of mixture}} = \frac{(m_C + m_H)_{\text{mix}}}{m_{\text{mix}}} \tag{47}
$$

In the flame gases, carbon is present in any unburned fuel and in CO and CO_2; while hydrogen is present in unburned fuel, H_2, H_2O, and OH. Summing the mass fractions of carbon and hydrogen associated with each species yields:

$$
\begin{aligned}
f = {} & Y_{\mathrm{C_2H_6}} \frac{2M_{\mathrm{C}}}{M_{\mathrm{C_2H_6}}} + Y_{\mathrm{CO}} \frac{M_{\mathrm{C}}}{M_{\mathrm{CO}}} + Y_{\mathrm{CO_2}} \frac{M_{\mathrm{C}}}{M_{\mathrm{CO_2}}} + Y_{\mathrm{C_2H_6}} \frac{3M_{\mathrm{H_2}}}{M_{\mathrm{C_2H_6}}} \\
& + Y_{\mathrm{H_2}} + Y_{\mathrm{H_2O}} \frac{M_{\mathrm{H_2}}}{M_{\mathrm{H_2O}}} + Y_{\mathrm{OH}} \frac{0.5M_{\mathrm{H_2}}}{M_{\mathrm{OH}}}
\end{aligned} \tag{48}
$$

Substituting for the mass fractions,

$$
Y_i = \chi_i M_i / M_{\text{mix}} \tag{49}
$$

$$
f = \chi_{\mathrm{C_2H_6}} \frac{M_{\mathrm{C_2H_6}}}{M_{\text{mix}}} \frac{2M_{\mathrm{C}}}{M_{\mathrm{C_2H_6}}} + \chi_{\mathrm{CO}} \frac{M_{\mathrm{CO}}}{M_{\text{mix}}} \frac{M_{\mathrm{C}}}{M_{\mathrm{CO}}} + \cdots \tag{50}
$$

$$
f = \frac{(2\chi_{\mathrm{C_2H_6}} + \chi_{\mathrm{CO}} + \chi_{\mathrm{CO_2}})M_{\mathrm{C}}}{M_{\text{mix}}} + \frac{(3\chi_{\mathrm{C_2H_6}} + \chi_{\mathrm{H_2}} + \chi_{\mathrm{H_2O}} + 0.5\chi_{\mathrm{OH}})M_{\mathrm{H_2}}}{M_{\text{mix}}} \tag{51}
$$

where,

$$
M_{\text{mix}} = \sum \chi_i M_i \tag{52}
$$

3.3 Simplified Analysis

3.3.1 Non-premixed turbulent flames with equilibrium chemistry:

The general pointers for this analysis are:

- Chemicals react to equilibrium as fast as they mix.
- Remaining question is how the fuel mixes with the oxidizer.
- All scalar diffusivities are equal.
- By computing the mixing of mixture fraction f, the mixing of everything can be computed.
- Conservation equation for the mixture fraction can be written.
- If one further assumes that energy diffuses at the same rate for all species, i.e., Lewis number $\text{Le} = k/(D\rho c_p) = 1$, and no heat transfer occurs, then the enthalpy field, as well as the temperature field, can be uniquely described by f.
- Thus, assuming:
  - equilibrium (fast) chemistry
  - equal diffusivity and Le=1
  - no heat loss
- all scalar variables (temperature, mass fractions, and density) are known functions of mixture fraction only.

3.3.2 Finite-rate chemistry in non-premixed flames:

The general pointers for this analysis are:

- If one wishes to relax the fast chemistry assumption, then, in addition to a conservation equation for total mass, energy, and momentum, each species will have a conservation equation with a chemical source term, $M_i\omega_i$. The source term is the sum of all chemical kinetic reactions that involve species i.
- These kinetic rates depend on other species and, more importantly, have a nonlinear dependence on both species and temperature.
- As the mixing rate increases, one chemical process will emerge at first to depart from chemical equilibrium.
- Increasing the mixing rate further will result in another process departing from equilibrium.
- One by one, processes will depart from equilibrium until the main energy releasing reactions are competing with the mixing rate.
- As the mixing rate increases further, the temperature begins to depart from equilibrium solution.

3.4 Flame Length

Many definitions and techniques for measuring flame lengths are found in the literature, and no single definition is accepted as preferred. Therefore, care must be exercised in comparing results of different investigators and in the application of correlation formulae. Common definitions of flame length include visual determinations by a trained observer, length based on the axial length at which mixture fraction is stoichiometric, and axial location of the average peak centerline temperature. In general, visible flame lengths tend to be larger than those based on temperature or concentration measurements. Therefore, care must be exercised in comparing results of different investigators and in the application of correlation formulae.

3.4.1 Factors affecting Flame length

For vertical flames created by a fuel jet issuing into a quiescent environment, four primary factors determine flame length:

- Relative importance of initial jet momentum flux and buoyant forces acting on the flame - Froude Number
- Stoichiometry - stoichiometric mixture fraction
- Ratio of the nozzle fluid (fuel) to ambient gas density - density ratio
- Initial jet diameter
- Gravity (buoyancy)

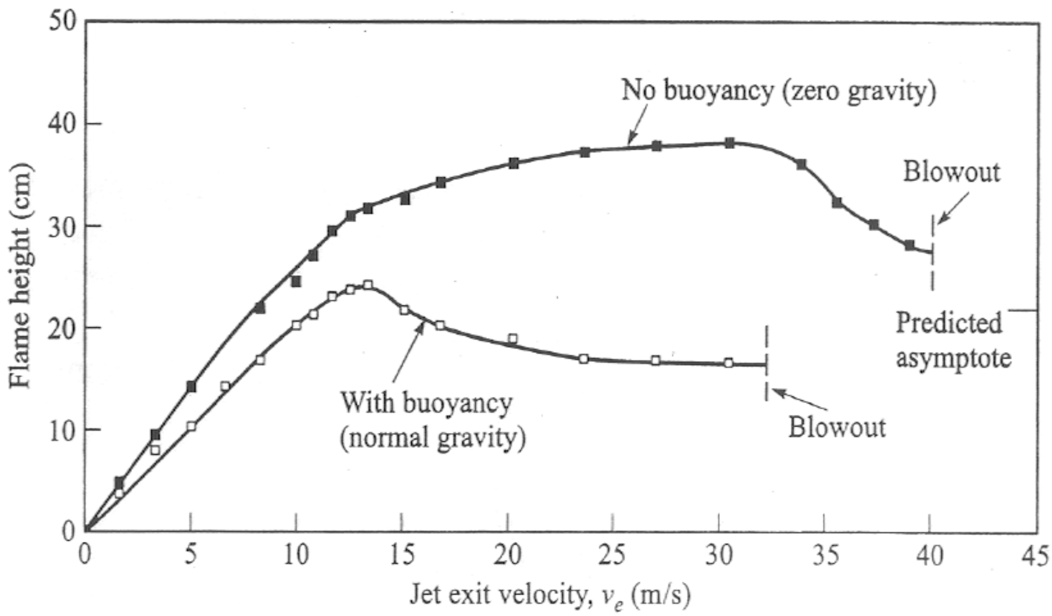


Figure 9: Comparison of jet flame heights with and without buoyancy

For turbulent jet flames, the density ratio and the jet diameter can be combined to give a momentum diameter, defined as:

$$d_j^\star = d_j(\rho_e/\rho_\infty)^{1/2} \tag{53}$$

The basic idea in the above definition is that jets with identical initial jet momentum fluxes should have identical velocity fields. Increasing the density of the nozzle fluid produces same effect as increasing the nozzle diameter. Defining a dimensionless flame height:

$$L^\star \equiv \frac{L_f f_s}{d_j(\rho_e/\rho_\infty)^{1/2}} = \frac{L_f f_s}{d_j^\star} \tag{54}$$

Now we define the following flame Froude number for turbulent jet flames,

$$\mathrm{Fr_f} = \frac{v_e f_s^{3/2}}{\left(\frac{\rho_e}{\rho_\infty}\right)^{1/4}\left[\frac{\Delta T_f}{T_\infty} g d_j\right]^{1/2}} \tag{55}$$

where ΔT_f is the characteristic temperature rise as a result of combustion. Based on the relative magnitudes of jet momentum to buoyancy, the jet flame exhibits different regimes: 1) $\mathrm{Fr_f} < 1$ where flames are dominated by buoyancy, and 2) $\mathrm{Fr_f} >> 1$ where flames are dominated by momentum.

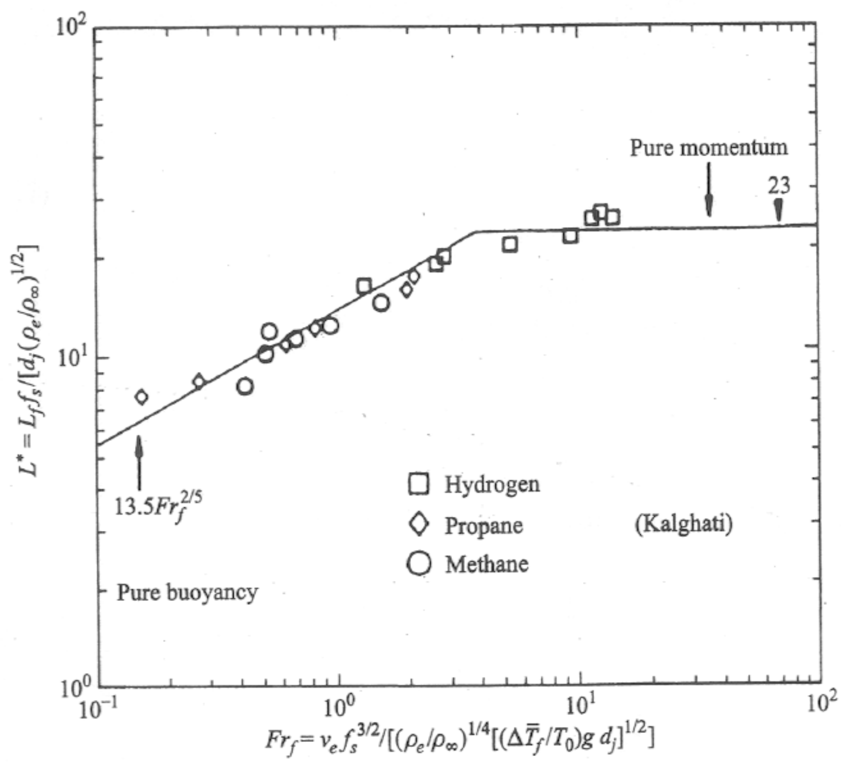


Figure 10: Flame lengths for jet flames correlated with flame Froude number

A buoyancy-dominated regime correlated by:

$$L^\star = \frac{13.5\mathrm{Fr_f^{2/5}}}{(1+0.07\mathrm{Fr_f^2})^{1/5}} \quad \text{for} \quad \mathrm{Fr_f} < 5 \tag{56}$$

and a momentum-dominated regime where the dimensionless flame length is constant:

$$L^\star = 23 \quad \text{for} \quad \mathrm{Fr_f} \geq 5 \tag{57}$$

Although various other correlations are available in literature, the one discussed above is one of the most used correlations for practical calculations.

3.5 Stability: Liftoff and Blowout of Non-premixed Flames:

A jet flame will lift from an attached position at the burner exit if the exit velocity is sufficiently high. The liftoff height, the distance between the burner tip and the base of the flame, will increase with additional increases in velocity until the flame blows out. Currently there is no consensus on what determines the liftoff height. However, the three mechanisms below are the most favored ones in literature:

- *Theory 1*: Local flow velocity at the base of the flame matches the turbulent premixed flame velocity corresponding to local conditions at the base of the flame.
- *Theory 2*: Local strain rates in the fluid at the base of the flame exceed the extinction strain rate for a laminar flame corresponding to local conditions at the base of the flame.
- *Theory 3*: The time available for backmixing by large-scale flow structures of hot products with fresh mixture is less than a critical chemical time required for ignition.

The following figure shows the liftoff heights for methane, propane, and ethylene jet flames as functions of initial jet velocity. Kalghatgi [58], interpreting his data in terms of Theory I, developed

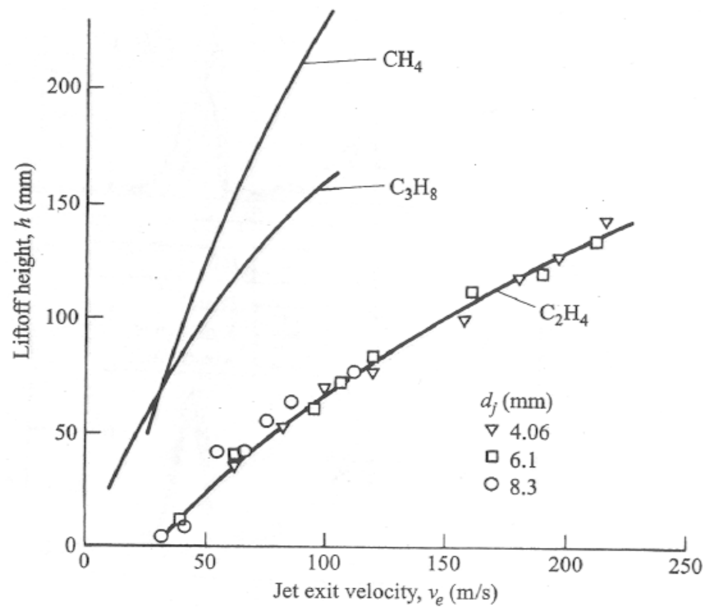


Figure 11: Jet flame liftoff heights versus jet exit velocity.

the following correlation that can be used to describe the liftoff behavior of hydrocarbon–air flames:

$$\frac{\rho_e S_{L,\max} h}{\mu_e} = 50 \left(\frac{v_e}{S_{L,\max}}\right)\left(\frac{\rho_e}{\rho_\infty}\right)^{1.5} \tag{58}$$

where $S_{L,\max}$ is the maximum laminar flame speed, which occurs near stoichiometric ratios for hydrocarbons. Based on the same theory, the following correlation was proposed by Kalghathi to estimate blowout flowrates for jet flames:

$$\frac{v_e}{S_{L,max}}\left(\frac{\rho_e}{\rho_\infty}\right)^{1.5} = 0.017 \cdot \mathrm{Re_H}(1 - 3.5 \cdot 10^{-6}\mathrm{Re_H}) \tag{59}$$

where the Reynolds number, $\mathrm{Re_H}$ is defined as:

$$\mathrm{Re_H} = \frac{\rho_e S_{L,max} H}{\mu_e} \tag{60}$$

The characteristic length, H, is the distance along the burner axis where the mean fuel concentration has decreased to its stoichiometric value and can be estimated by:

$$H = 4\Big[\frac{Y_{F,e}}{Y_{F,st}}\left(\frac{\rho_e}{\rho_\infty}\right)^{1/2} - 5.8\Big] d_j \tag{61}$$

The fig.12 demonstrates the applicability of Eq.(10) to a wide range of fuels.

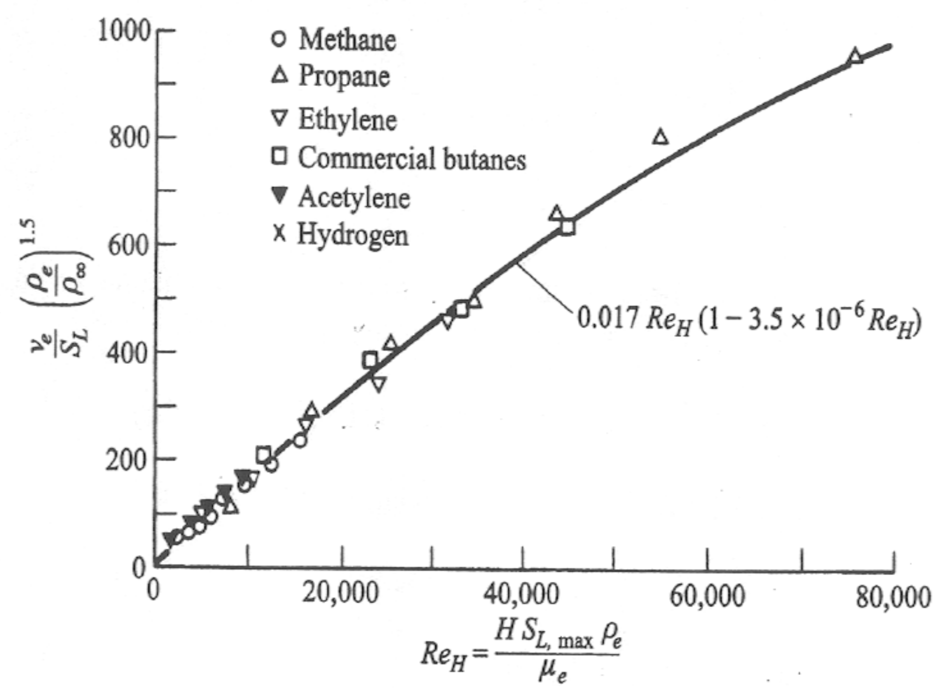


Figure 12: Universal Stability Blowout curve